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M2

Started on	Saturday, 12 September 2026, 9:44 PM
State	Finished
Completed on	Saturday, 12 September 2026, 9:45 PM
Time taken	1 min 18 secs
Marks	1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findFloor(int arr[], int low, int high, int x)
4  {
5      if (low > high)
6          return -1;
7
8      int mid = (low + high) / 2;
9
10     if (arr[mid] == x)
11         return arr[mid];
12
13     if (arr[mid] < x)
14     {
15         int ans = findFloor(arr, mid + 1, high, x);
16
17         if (ans == -1)
18             return arr[mid];
19
20         return ans;
21     }
22
23     return findFloor(arr, low, mid - 1, x);
24 }
25
26 int main()
27 {
28     int n, x;
29
30     scanf("%d", &n);
31
32     int arr[n];
33
34     for (int i = 0; i < n; i++)
35         scanf("%d", &arr[i]);
36
37     scanf("%d", &x);
38
39     printf("%d", findFloor(arr, 0, n - 1, x));
40
41     return 0;
42 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)